WORKED SOLUTION

STEP I 1988, Question 6

Pure • Complex Numbers and Geometry

STEP I 1988, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q6

The key idea

In the Argand diagram the vector $\overrightarrow{P_jP_k}$ is just the complex number $z_k - z_j$, so equal vectors become equal differences, and perpendicularity becomes "one difference divided by the other is purely imaginary". Once the given values are substituted, condition (i) pins down z_4 and w in terms of z, and condition (ii) fixes the imaginary part of z. The hexagon then turns out to be symmetric about its centre, which makes the convexity and the final angle easy to handle.

Solution

Translating the statements

Since $\overrightarrow{P_jP_k}$ is represented by $z_k - z_j$, statement (i) says

$$z_2 - z_1 = z_4 - z_5 \qquad \text{and} \qquad z_3 - z_2 = z_5 - z_6.$$

For (ii), two non-zero vectors u and v are perpendicular exactly when their arguments differ by $\pm\frac{\pi}{2}$, that is, when u/v is purely imaginary. The points are distinct, so $z_4 - z_2 \neq 0$ and $z_6 - z_3 \neq 0$, and statement (ii) says

$$\frac{z_4 - z_2}{z_6 - z_3} \text{ is purely imaginary,} \qquad \text{equivalently} \qquad \operatorname{Re}\left[(z_4 - z_2)\,\overline{(z_6 - z_3)}\right] = 0.$$

The consequence of (i)

Adding the two equations in (i), the z_2 and z_5 terms cancel:

$$z_3 - z_1 = z_4 - z_6, \qquad \text{so} \qquad z_4 - z_3 = z_6 - z_1,$$

which is the statement $\overrightarrow{P_3P_4} = \overrightarrow{P_1P_6}$, as required.

The hexagon

Now put $z_1 = 0$, $z_2 = 1$, $z_3 = z$, $z_5 = \mathrm{i}$ and $z_6 = w$. The first equation of (i) gives

$$z_4 = z_5 + z_2 - z_1 = 1 + \mathrm{i},$$

and the second gives $z - 1 = \mathrm{i} - w$, so

$$w = 1 + \mathrm{i} - z.$$

Notice that $z_1 + z_4 = z_2 + z_5 = z_3 + z_6 = 1 + \mathrm{i}$: opposite vertices are symmetric about the point $\frac{1}{2}(1 + \mathrm{i})$, so the hexagon has a centre of symmetry.

For (ii), $z_4 - z_2 = \mathrm{i}$ and $z_6 - z_3 = w - z = 1 + \mathrm{i} - 2z$. The quotient $\mathrm{i}/(1 + \mathrm{i} - 2z)$ is purely imaginary exactly when $1 + \mathrm{i} - 2z$ is real, that is, when $1 - 2\,\mathrm{Im}(z) = 0$. So

$$\mathrm{Im}(z) = \tfrac{1}{2}, \qquad \text{and then} \qquad \mathrm{Im}(w) = 1 - \tfrac{1}{2} = \tfrac{1}{2}.$$

Write $z = a + \frac{1}{2}\mathrm{i}$ with a real. Then $w = (1 - a) + \frac{1}{2}\mathrm{i}$, so $\mathrm{Re}(w) = 1 - \mathrm{Re}(z)$ always. The six vertices are

$$(0,0), \quad (1,0), \quad (a, \tfrac{1}{2}), \quad (1,1), \quad (0,1), \quad (1-a, \tfrac{1}{2}) :$$

the unit square with a "point" added halfway up each vertical side.

Ask yourself How can we test convexity without relying on a picture?

Walk round the hexagon in order. It is convex exactly when we always turn the same way at each vertex. The turn at a vertex is measured by the sign of the cross product $\mathrm{Im}(\bar{u}v) = u_x v_y - u_y v_x$ of the incoming edge u and the outgoing edge v. The edges are

$$P_1P_2 = (1,0), \quad P_2P_3 = (a-1, \tfrac{1}{2}), \quad P_3P_4 = (1-a, \tfrac{1}{2}),$$
$$P_4P_5 = (-1,0), \quad P_5P_6 = (1-a, -\tfrac{1}{2}), \quad P_6P_1 = (a-1, -\tfrac{1}{2}),$$

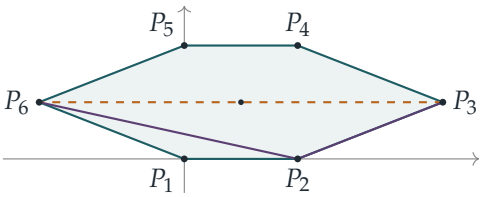
and the cross products at $P_2, P_3, P_4, P_5, P_6, P_1$ in turn are

$$\tfrac{1}{2}, \qquad a - 1, \qquad \tfrac{1}{2}, \qquad \tfrac{1}{2}, \qquad a - 1, \qquad \tfrac{1}{2}.$$

Four of these are positive whatever a is, so the hexagon is convex if and only if $a - 1 > 0$. (If $a = 1$, then P_3 lies on the segment P_2P_4 and we do not have a genuine hexagon.) Geometrically, P_3 must lie to the right of the side $x = 1$ of the square and, by the symmetry, P_6 to the left of $x = 0$. So the conditions are

$$\boxed{\mathrm{Re}(z) > 1 \quad \text{and} \quad \mathrm{Re}(w) = 1 - \mathrm{Re}(z) \ \ (\text{so } \mathrm{Re}(w) < 0)}$$

(together with $\mathrm{Im}(z) = \mathrm{Im}(w) = \frac{1}{2}$, which (ii) already forces).



The distance P_3P_6

At P_2 the two arms are

$$z_3 - z_2 = (a-1) + \tfrac{1}{2}\mathrm{i}, \qquad z_6 - z_2 = -a + \tfrac{1}{2}\mathrm{i}.$$

For the angle $\theta = \angle P_3P_2P_6$ (between 0 and π), the dot product of the arms gives $|P_2P_3|\,|P_2P_6|\cos\theta$, and the size of the cross product gives $|P_2P_3|\,|P_2P_6|\sin\theta$:

$$\text{dot} = -a(a-1) + \tfrac{1}{4} = \tfrac{1}{4}(1 + 4a - 4a^2), \qquad \text{cross} = (a-1)\cdot\tfrac{1}{2} - \tfrac{1}{2}\cdot(-a) = \tfrac{1}{2}(2a-1).$$

With $a > 1$ the cross product is positive, so it is exactly the $\sin\theta$ term, and

$$\tan\theta = \frac{\frac{1}{2}(2a-1)}{\frac{1}{4}(1+4a-4a^2)} = \frac{2(2a-1)}{1+4a-4a^2}.$$

Setting this equal to $-\frac{2}{3}$:

$$3(2a-1) = -(1+4a-4a^2) \quad\Longrightarrow\quad 4a^2 - 10a + 2 = 0 \quad\Longrightarrow\quad 2a^2 - 5a + 1 = 0,$$

so $a = \frac{1}{4}(5 \pm \sqrt{17})$. The minus sign gives $a \approx 0.22$, which breaks the convexity condition $a > 1$; the plus sign gives $a \approx 2.28 > 1$. (There the denominator $1 + 4a - 4a^2$ is negative, so θ is obtuse, as a negative tangent requires.)

Finally P_3 and P_6 have the same imaginary part, so

$$P_3P_6 = |z - w| = |2a - 1| = 2a - 1 = \frac{5+\sqrt{17}}{2} - 1 = \boxed{\frac{3+\sqrt{17}}{2}} \approx 3.56.$$

If we drop the convexity requirement, the same method (now with $|2a-1|$ in the numerator) gives one further solution, $a = \frac{1}{4}(-1 - \sqrt{17})$. This is just the mirror image $a \mapsto 1 - a$, which swaps z and w, and it gives the same distance $\frac{1}{2}(3 + \sqrt{17})$.

Checks

- Symmetry: $z_1 + z_4 = z_2 + z_5 = z_3 + z_6$, so the result $\overrightarrow{P_3P_4} = \overrightarrow{P_1P_6}$ is just the third pair of opposite sides of a centrally symmetric hexagon.
- With $a = \frac{1}{4}(5 + \sqrt{17}) \approx 2.2808$, a direct computation of $\arg\big((z_6 - z_2)/(z_3 - z_2)\big)$ gives an angle whose tangent is -0.66667, and $|z - w| \approx 3.5616 = \frac{1}{2}(3 + \sqrt{17})$.
- Testing convexity numerically for a from -5 to 5 confirms that the hexagon is convex exactly when $a > 1$.

Common slips

- Getting the direction of a vector wrong: $\overrightarrow{P_5P_4}$ is $z_4 - z_5$, not $z_5 - z_4$.

- Writing perpendicularity as "the product of gradients is -1" and then losing the vertical case. Here $\overrightarrow{P_2P_4}$ is vertical, and the quotient test handles this cleanly.
- Keeping the root $a = \frac{1}{4}(5 - \sqrt{17})$, which does not give a convex hexagon (and in fact gives $\tan\theta > 0$ there).

Beyond STEP

Treating points as complex numbers turns geometry into algebra: translations are additions, and rotations and scalings are multiplications. The test "u/v is purely imaginary" is one example of how angles become arguments. The cross-product sign test for convexity is the basis of the standard convex-hull algorithms in computational geometry. Needham's *Visual Complex Analysis* develops this geometric view of complex numbers beautifully.

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